

Alaa Eldin Abdelaal

The Collaborative Haptics and Robotics in Medicine (CHARM) Lab, Mechanical Engineering Department, Stanford University, Building 660-126, 418 Panama Mall, Stanford CA 94305-2203

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EDUCATION

University of British Columbia Vancouver, Canada
Ph.D., in Electrical and Computer Engineering September 2017 – December 2022
Advisors: Prof. Tim Salcudean and Prof. Gregory Hager (at Johns Hopkins University)

Simon Fraser University Burnaby, Canada
M.Sc., in Computing Science September 2015 – August 2017
Thesis: Event-based Control as a Cloud Service
Advisor: Prof. Mohamed Hefeeda
Cumulative GPA: 4.0

Mansoura University Mansoura, Egypt
B.Sc., in Computers and Systems Engineering September 2007 - June 2012
Distinction with honor degree, Total grade: (90.68% - ranked 1st)
Thesis/Graduation Project: One-Eyed Self-Learning Arm Robot.

ACADEMIC HONORS & AWARDS

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- Stanford Emerging Technology Review Fellow, at Stanford University 2023 - 2025
 - Medical Robotics Pioneer: I was selected as one of two rising stars in medical robotics during a workshop at the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023). October 2023
 - RSS Pioneer: I was selected to participate in the Robotics: Science and Systems (RSS) Pioneers Workshop, a highly selective workshop, bringing together a cohort of the world's top early career researchers in robotics. 2023
 - NSERC Postdoctoral Fellowship, held at Stanford University: my application received the highest ranking in Computer Science in Canada (\$90,000 CAD) January 2023 – December 2024
 - Canada Graduate Scholarships – Michael Smith Foreign Study Supplements (\$6,000 CAD) Spring 2022
 - Vanier Canada Graduate Scholarship, the most prestigious scholarship for PhD students in Canada (\$50,000 CAD per year for three years) 2019 - 2022
 - President's Academic Excellence Initiative PhD Award at the University of British Columbia 2020 – 2022
 - HRI Pioneer: I was selected to participate in the Human-Robot Interaction (HRI) Pioneers Workshop, a highly selective workshop, bringing together a cohort of the world's top student researchers in this area. 2020
 - ACM SIGCHI travel award to attend the ACM/IEEE International Conference on Human-Robot Interaction (HRI) in Cambridge, UK. 2020
 - Best Bench-to-Bedside Paper Award at the 10th International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), Rennes, France 2019
 - The IEEE Robotics and Automation Society Travel Grant to attend the International Conference on Robotics and Automation (ICRA) 2019
 - The Faculty of Applied Science Graduate Award at the University of British Columbia, three times 2017, 2018 and 2022
 - The International Tuition Award at the University of British Columbia 2017 - 2019
 - The School of Computing Science Travel Award at Simon Fraser University Summer 2017
 - The School of Computing Science Graduate Fellowship at Simon Fraser University Fall 2015
 - Graduate Scholarship at Carnegie Mellon University (\$42,750 USD per year) (declined) Fall 2015
 - Laboratory Instruction Graduate Fellowship at the American University in Cairo (declined) Spring 2015
 - Sponsored Research Tuition Award at the University of Louisville (\$22,742 USD) (declined) 08/2014 – 07/2015
 - Distinguished Graduate Award from the Egyptian Engineers Syndicate 2012
 - My graduation project was one of the winning projects in the 6th Annual Egypt's Young Entrepreneurs Competition of Injaz Egypt and we received a seed fund of 60,000 Egyptian pounds from ExxonMobil Egypt 2012
 - The Young Innovator award from Nahdet El Mahrousa Association and a prize of 6,000 Egyptian Pounds, which is given to the best bachelor theses in Egypt 2012
 - Distinction Award from the Faculty of Engineering, Mansoura University, Egypt, five times 2007 – 2012

RESEARCH EXPERIENCE

Stanford University Stanford, CA, USA
Mechanical Engineering Dept. February 2023 – Present

NSERC Postdoctoral Scholar

Working with Prof. Allison Okamura at the Collaborative Haptics in Robotics and Medicine Laboratory. Collaborating with Prof. Jeannette Bohg. Conducting research in robot learning and robotic surgery.

University of British Columbia

Electrical and Computer Engineering Dept.

Research Assistant

Working with Prof. Tim Salcudean at the Robotics and Control Lab and Prof. Gregory Hager at Johns Hopkins University. Conducting research at the intersection of Autonomy and Human-Robot Interaction with application to surgical robotics.

Vancouver, BC, Canada

September 2017 – December 2022

Johns Hopkins University

Computer Science Dept.

Visiting Graduate Scholar

Working with Prof. Gregory Hager at Computational Interaction and Robotics Lab (CIRL), Prof. Masaru Ishii at Johns Hopkins School of Medicine and Prof. Tim Salcudean at UBC. Conducting research in the area of surgical training and skill assessment in minimally invasive surgery.

Baltimore, MD, USA

July 2021– June 2022

Simon Fraser University

School of Computing Science

Research Assistant

Working with Prof. Mohamed Hefeeda at the Network Systems Lab. Conducting research in the area of Automation and Control systems, Cloud Computing and Distributed Systems.

Burnaby, BC, Canada

March 2015– August 2017

JOURNAL ARTICLES

- Nigel Nelson, Juo-Tung Chen, Jesse Haworth, Xinhao Chen, Lukas Zbinden, Dianye Huang, **Alaa Eldin Abdelaal**, et al. “*Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics*”. Submitted to Science Robotics 2026.
- Zhaoyang Jacopo Hu, Alex Ranne, **Alaa Eldin Abdelaal**, Etienne Burdet, Allison M. Okamura, Ferdinando Rodriguez y Baena. “*How are my stitches? Model selection and real-time skill assessment for suturing in robotic surgery*”. Submitted 2026.
- **Alaa Eldin Abdelaal**, Jiaying Fang, Tim N. Reinhart, Jacob A. Mejia, Tony Z. Zhao, Jeannette Bohg, Allison Okamura, “*Force-Aware Autonomous Robotic Surgery*”. Submitted 2025.
- Alexandre Banks, Randy Moore, Sayem Zaman, **Alaa Eldin Abdelaal**, Septimiu E. Salcudean “*AutoCam: Hierarchical Path Planning for an Autonomous Auxiliary Camera in Surgical Robotics*”. In IEEE Transactions on Medical Robotics and Bionics, Minor Revision, 2025.
- **Alaa Eldin Abdelaal**, Rachelle Van Rumpt, Sayem Zaman, Irene Tong, Anthony Jarc, Gary L. Gallia, Masaru Ishii, Gregory D. Hager, Septimiu E. Salcudean. “*The Quiet Eye Phenomenon in Minimally Invasive Surgery*”. In the International Journal of Computer Assisted Radiology and Surgery, 20:2087–2093, 2025.
- Alexandre Banks, **Alaa Eldin Abdelaal**, Septimiu E. Salcudean, “*Head Motion-Corrected Eye Gaze Tracking with the da Vinci Surgical System*”. In the International Journal of Computer Assisted Radiology and Surgery, pp. 1459–1467, 2024.
- Kenneth Perrone, **Alaa Eldin Abdelaal**, Carla Pugh, Allison Okamura, “*Haptics: The Science of Touch as a Foundational Pathway to Precision Medical Education and Assessment*”. In Academic Medicine, pp. S84–S88, 2024.
- **Alaa Eldin Abdelaal**, Jordan Liu, Nancy Hong, Gregory D. Hager, Septimiu E. Salcudean. “*Parallelism in Autonomous Robotic Surgery*”. In IEEE Robotics and Automation Letters, 6.2:1824–1831, 2021. [Also presented at the IEEE International Conference on Robotics and Automation (ICRA), Xi'an, China, May 2021].
- **Alaa Eldin Abdelaal**, Apeksha Avinash, Megha Kalia, Gregory D. Hager, Septimiu E. Salcudean. “*A multi-camera, multi-view system for training and skill assessment for robot-assisted surgery*”. In the International Journal of Computer Assisted Radiology and Surgery, 15:1369–1377, 2020.
- **Alaa Eldin Abdelaal**, Prateek Mathur, Septimiu E. Salcudean. “*Robotics in Vivo: A Perspective on Human-Robot Interaction in Surgical Robotics*”. In Annual Review of Control, Robotics, and Autonomous Systems, 3:221–242, 2020.
- Apeksha Avinash, **Alaa Eldin Abdelaal**, Prateek Mathur, Septimiu E. Salcudean. “*A “pickup” stereoscopic camera with visual-motor aligned control for the da Vinci surgical system: a preliminary study*”. In the International Journal of Computer Assisted Radiology and Surgery, 14.7:1197–1206, 2019. [Winner of Best Bench-to-Bedside Award at the 10th International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), Rennes, France 2019].
- **Alaa Eldin Abdelaal**, Maram Sakr, Apeksha Avinash, Shahed K. Mohammed, Armaan. K. Bajwa, Mohakta Sahni, Soheil Hor, Sideny Fels, Septimiu E. Salcudean. “*Play Me Back: A Unified Training Platform for Robotic and Laparoscopic Surgery*”. In IEEE Robotics and Automation Letters, 4.2:554–561, 2018. [Also presented at the IEEE International Conference on Robotics and Automation (ICRA), Montreal, QC, Canada, May 2019].

MANUSCRIPTS IN PREPERATION

- Fredrik Samdal Solberg, Soham Sinha, Gadiel Sznaiers Camps, **Alaa Eldin Abdelaal**, Allison M. Okamura, Mac Schwager, Mark Skylar-Scott. “*Truly Omnidirectional 3D Bioprinting*”. In preparation, to be submitted to Nature, 2026.
- **Alaa Eldin Abdelaal**, Minahil Raza, Nada Elmasry, Allison M. Okamura. “A Multimodal Framework for Force Estimation in Robot-Assisted Surgery”. In preparation, 2026.
- **Alaa Eldin Abdelaal**, Zhuohong He, Chetan Reddy Narayanaswamy, Jiaqi Shao, Howard Ji, Allison M. Okamura. “*Human–Robot Interaction in Trilateral Manipulation for Autonomous Robotic Surgery*”. In preparation, 2026.

PEER-REVIEWED CONFERENCE PUBLICATIONS

- **Alaa Eldin Abdelaal**, Nancy Hong, Apeksha Avinash, Divya Budihal, Maram Sakr, Gregory D. Hager, Septimiu E. Salcudean, “*Orientation Matters: 6-DoF Autonomous Camera Movement for Video-based Skill Assessment in Robot-Assisted Surgery*”. In Proc. of the IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob), Seoul, Korea, August 2022.
- Apeksha Avinash, **Alaa Eldin Abdelaal**, Septimiu E. Salcudean. “*Evaluation of Increasing Camera Baseline on Depth Perception in Surgical Robotics*”. In Proc. of the IEEE International Conference on Robotics and Automation (ICRA), Paris, France, May 2020.
- **Alaa Eldin Abdelaal**, Tamir Hegazy, Mohamed Hefeeda. “*Event-based Control as a Cloud Service*”. In Proc. of the American Control Conference (ACC), Seattle, WA, May 2017.
- **Alaa Eldin Abdelaal**, Maram Sakr, Richard Vaughan. “*LOST Highway: a Multiple-Lane Ant-Trail Algorithm to Reduce Congestion in Large-Population Multi-Robot Systems*”. In Proc. of the 14th Conference on Computer and Robot Vision (CRV), Edmonton, Alberta, May 2017.

PEER-REVIEWED WORKSHOP PUBLICATIONS

- **Alaa Eldin Abdelaal**, Gregory D. Hager, Septimiu E. Salcudean. “*Multimodal Training by Demonstration for Robot-Assisted Surgery*.” In Proc. of the ACM/IEEE International Conference on Human-Robot Interaction (HRI), HRI Pioneers Workshop, Cambridge, UK, March 2020. (Acceptance rate: ~24%)

PATENTS

- Allison M. Okamura, Brian Vuong, Trevor Perey, Ariane Richard, **Alaa Eldin Abdelaal**, Ghadi Nehme, Antonio Meola. “*Robotic Stereotactic System for Computer-Assisted Brain Microsurgery*”. U.S. Provisional Patent Application, 2025
- Septimiu E. Salcudean, Omid Mohareri, Prateek Mathur, Apeksha Avinash, **Alaa Eldin Abdelaal**. “*Imaging device for use with surgical instrument*”. U.S. Patent Application No 16/862,371, 2020.

THESES

- **Alaa Eldin Abdelaal**, “*Human Skill Augmentation in Robot-Assisted Surgery*”. Ph.D. Thesis, Electrical and Computer Engineering Department, University of British Columbia, Vancouver, BC, Canada. December 2022.
- **Alaa Eldin Abdelaal**, “*Event-based Control as a Cloud Service*”. M.Sc. Thesis, School of Computing Science, Faculty of Applied Sciences, Simon Fraser University, Burnaby, BC, Canada. April 2/017.
- **Alaa Eldin Abdelaal et. al**, “*One-Eyed Self-Learning Arm Robot*”. B.Sc. Thesis, Computers and systems Engineering Department, Faculty of Engineering, Mansoura University, Mansoura, Egypt. June 2012.

GRANTS

Seed Grant from the Stanford Institute for Human-Centered Artificial Intelligence (HAI)	January - December 2025
Co-Principal Investigator with Profs. Allison Okamura, Carla Pugh and Jeannette Bohg	\$75,000 USD
Title: “ <i>Towards Collaborative Autonomous Surgical Assistant: A Case Study on Automating an Auxiliary Robotic Arm in Robot-Assisted Surgery</i> ”	

Seed Grant from the Stanford Institute for Human-Centered Artificial Intelligence (HAI)	January - December 2023
Co-Principal Investigator with Profs. Allison Okamura, Carla Pugh and Jeannette Bohg	\$75,000 USD
Title: “ <i>Force-based Robot Learning from Demonstration for Soft Tissue Manipulation</i> ”	

Intuitive Technology Research Grant, from Intuitive Surgical Inc.	January - December 2023
Co-Principal Investigator with Profs. Allison Okamura, Carla Pugh and Jeannette Bohg	\$60,000 USD
Title: “ <i>Towards Understanding the Role of Tool-Tissue Interaction Forces in Autonomous Robotic Surgery</i> ”	

Intuitive Foundation Clinical Research Grant, a subsidiary from Intuitive Surgical Inc.	August 2021 - July 2022
Co-Principal Investigator with Profs. Tim Salcudean, Gregory Hager and Masaru Ishii	\$60,000 USD
Title: “ <i>Quiet Eye Training in Minimally Invasive Surgery</i> ”	

REPORTS

- **Alaa Eldin Abdelaal**, Cosima du Pasquier, Brian Vuong, Allison M. Okamura. “Stanford Emerging Technology Review 2025: Robotics Chapter”. A policy-oriented report published by The Hoover Institution at Stanford University to inform government and industry leaders on advances in robotics and AI, 2025.
- **Alaa Eldin Abdelaal**, and Allison M. Okamura. “Stanford Emerging Technology Review 2024: Robotics Chapter”. A policy-oriented report published by The Hoover Institution at Stanford University to inform government and industry leaders on advances in robotics and AI, 2024.

TEACHING EXPERIENCE

State University of New York (SUNY) Upstate Medical University

Guest Lecturer

Discovery Science to Patient Care

Syracuse, New York, USA

February 2024

Stanford University

Bioengineering Dept.

Guest Lecturer

BIOE 301B: Clinical Needs and Technology

Stanford, CA, USA

March 2023

University of Prince Edward Island, Cairo Campus

Faculty of Sustainable Design

Guest Lecturer

ENG-4330: Innovations in Biomedical Engineering

Cairo, Egypt

October 2021

University of British Columbia

Electrical and Computer Engineering Dept.

Teaching Assistant

EECE 589/MECH 464/MECH 563: Introduction to Robotics

Vancouver, BC Canada

January 2021 – April 2021

Simon Fraser University

School of Computing Science

Teaching Assistant

CMPT376W: Technical Writing and Group Dynamics

Burnaby, BC Canada

May 2017 – August 2017

Mansoura University

Computers and Systems Engineering Dept.

Teaching Assistant

CSE 3116: Introduction to Control Engineering, CSE 3413: Machine Learning, and MUR 112: Introduction to Programming

Mansoura, Egypt

September 2012 – February 2015

MENTORSHIP

- Howard Ji, Undergraduate student, Computer Science, Stanford University July 2025 – December 2025
- Jiaqi Shao, MS student, Mechanical Engineering, Stanford University April 2025 – December 2025
- Chetan Narayanaswamy, MS student, Mechanical Engineering, Stanford University January 2025 – December 2025
- Joonwon Kang, MS student, Mechanical Engineering, Stanford University January 2025 – March 2025
- Amber Xie, PhD student, Computer Science, Stanford University April 2024 – June 2024
- Tim Niklaus Reinhart, MS student, Computer Science, Stanford University January 2024 – June 2024
- Jiaying Fang, MS student, Electrical Engineering, Stanford University September 2023 – December 2023
- Jacob A. Mejia, MS student, Computer Science, Stanford University June 2023 – August 2023
- Mihir Rao, Undergraduate student, EECS, UC Berkeley May 2023 – July 2023
- Mary Kate Gale, PhD student, Bioengineering, Stanford University January 2023 – May 2023
- Brian Vuong, PhD student, Mechanical Engineering, Stanford University January 2023 – May 2023
- Paula Stocco, MSc student, Mechanical Engineering, Stanford University January 2023 – March 2023
- Huilin (Lin) Xu, Undergraduate student, Computer Science, Johns Hopkins University May 2022 – December 2022
- Topic: “A Multi-Modal, Multi-Institutional Data Set for Autonomous Robotic Surgery”
- Rachelle Van Rump, Undergraduate student, ECE, UBC January 2022 – August 2022
- Topic: “Quiet Eye Training in Minimally-Invasive Surgery”
- Sayem Zaman, Undergraduate student, Engineering Physics, UBC May 2021 – August 2022
- Topic: “Gaze-based Learning from Demonstration for Autonomous Robotic Surgery”
- Isabelle Andre, Undergraduate student, ECE, UBC May 2021 – December 2021
- Topic: “Integrated Task and Motion Planning for Parallelism in Autonomous Robotic Surgery”

- Kritika Joshi, Undergraduate student, Engineering Physics, UBC
Topic: "Trilateral Manipulation in Autonomous Robotic Surgery" January 2021 – August 2021
- Jason Lee, Undergraduate student, Computer Science, UBC
Topic: "An Optimization-based Method for Camera Viewpoint Automation in Robotic Surgery" May 2021 - August 2021
- Kareem Alaswad, Undergraduate student, Mechanical Engineering, UBC
Topic: "A Simulation Interface for the da Vinci Surgical Robot" May 2021 - September 2021
- Nancy Hong, Undergraduate student, ECE, UBC
Topic: "Optimization-based Viewpoint Guidance for a 6-DoF Endoscope" May 2020 - March 2021
- Jordan Liu, Undergraduate student, Computer Science, UBC
Topic: "Parallelism in Autonomous Robotic Surgery" May 2020 - September 2020
- Linghao Lyu, Undergraduate student, ECE, UBC
Topic: "Eye gaze Tracking for Robot-Assisted Surgery" May 2019 - December 2020
- Divya Budihal, Undergraduate student, ECE, UBC
Topic: "Autonomous Surgical Camera Movement in 3D: A Learning from Demonstration Approach" September 2019 - April 2020
- Apeksha Avinash, MSc student, ECE, UBC
Topic: "A Stereoscopic "Pickup" Camera for the da Vinci Surgical System" May 2018 – January 2020

INVITED TALKS

- "Towards Force-Aware Autonomous Robotic Surgery". The 5th annual Stanford AI+HEALTH Conference, Stanford, CA, USA, December 2025.
- "Many Robots Make Light Work: Two Studies in Surgical Robotics and 3D Bioprinting". Johnson & Johnson MedTech, Santa Clara, CA, USA, October 2025.
- "Towards Force-Aware Autonomous Robotic Surgery". The 2025 International Symposium on Medical Robotics Workshop on Pushing the boundaries: safe and intelligent medical robotics. Atlanta, GA, USA, May 2025.
- "Human Skill Augmentation in Robot-Assisted Surgery"
 - Hamad Bin Khalifa University, the Mechanical Engineering Dept., Doha, Qatar, September 2025.
 - The Hybrid Human-Machine Interaction in Surgery Workshop at the Hamlyn Symposium on Medical Robotics, London, UK, June 2024.
 - Carnegie Mellon University, the Electrical and Computer Engineering Dept., Pittsburgh, Pennsylvania, March 2024.
 - The University of Illinois Chicago, the Dept. of Electrical and Computer Engineering, Chicago, Illinois, February 2024.
 - The University of Texas at San Antonio, the Dept. of Mechanical Engineering, San Antonio, Texas, February 2024.
 - Houston Methodist Institute for Robotics, Imaging & Navigation Symposium, Houston, Texas, USA, October 2023.
 - University of Alberta, the Electrical and Computer Engineering Department, Edmonton, Alberta, Canada, June 2023.
 - Queensland University of Technology, School of Electrical Engineering and Robotics, Brisbane, Australia, May 2022.
 - University of California, Santa Cruz, the Electrical and Computer Engineering Dept., Santa Cruz, California, May 2022.
 - Carnegie Mellon University, the Robotics Institute, Pittsburgh, Pennsylvania, USA, March 2022.
 - Rice University, the Electrical and Computer Engineering Department, Houston, Texas, USA, March 2022.
- "Parallelism in Autonomous Robotic Surgery". The 2022 IEEE International Conference on Robotics and Automation (ICRA) Workshop on Intelligent Control Methods and Machine Learning Algorithms for Human-Robot Interaction and Assistive Robotics, Philadelphia, Pennsylvania, USA, May 2022.
- "An "Additional View" on Human-Robot Interaction and Autonomy in Robot-Assisted Surgery". Disney Research, Los Angeles, USA, March 2022.
- "An "Additional View" on Human-Robot Interaction and Autonomy in Robot-Assisted Surgery". Johns Hopkins University, Baltimore, Maryland, USA, November 2021.
- "An "Additional View" on Human-Robot Interaction and Autonomy in Robot-Assisted Surgery". Cornell University, Ithaca, New York, USA, November 2020.
- "Training by Demonstration for Robot-Assisted Surgery". Intuitive Surgical Research Symposium, Sunnyvale, CA, USA, January 2020.

SERVICE

- Program committee member at the Robotics: Science and Systems (RSS) Pioneers Workshop which was held in conjunction with the 2024 RSS Conference.
- Panel Chair at the 16th annual Human-Robot Interaction (HRI) Pioneers Workshop which was held in conjunction with the 2021 ACM/IEEE International Conference on Human-Robot Interaction.
- Program committee member at the 11th IEEE International Conference on Healthcare Informatics (ICHI) 2023
- Reviewer at:
 - IEEE Transactions on Robotics (2026)
 - ACM Transactions on Human-Robot Interaction (2025)
 - npj Digital Medicine (2022)
 - IEEE/ASME Transactions on Mechatronics (2021)
 - IEEE Transactions on Control Systems Technology (2019)

- IEEE Robotics and Automation Letters (2020, 2021, 2022, 2023, 2024, 2025)
- The International Journal of Computer Assisted Radiology and Surgery (2021)
- IEEE Control Systems Letters (2020)
- The International Conference on Robotics and Automation (ICRA) (2021, 2022, 2023, 2024)
- The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2021, 2022, 2023, 2025)
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) (2020, 2021, 2022, 2023)
- The IEEE International Conference on Automation Science and Engineering (CASE) 2021
- International Conference for Biomedical Robotics and Bio-mechatronics (BioRob) 2020
- The 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) 2020.
- The American Control Conference (ACC) (2018, 2021, 2022)
- The International Symposium on Medical Robotics (ISMR) 2023
- MICCAI 2019 Workshop on Augmented Environments for Computer Assisted Interventions
- Poster session committee member at the Grace Hopper Celebration of Women in Computing Conference (GHC) 2019.
- Volunteer team leader at the organization of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2017).
- Volunteer at the organization of ROSCon 2017 (the main conference for the developers of Robot Operating System (ROS)).

ADDITIONAL TRAINING

Stanford University

Stanford, CA, USA

Preparing for Faculty Careers

April - June 2023

An intensive course on how to obtain a faculty job and thrive as a professor.

Imperial College London

London, The United Kingdom

The Hamlyn Winter School on Surgical Imaging and Vision

December 2019

The Hamlyn Winter School focuses on both the technical and clinical aspects of Surgical Imaging and Vision. Through invited lectures, hands-on demonstrations, workshops, and mini-projects, the purpose of this winter school is to help researchers familiarize with the cutting edge research of this field.

University of British Columbia

Vancouver, Canada

The Centre for the Integration of Research, Teaching and Learning (CIRTL) Summer Institute

Summer 2018

This summer institute was a five-day, hands-on, intensive teaching course that combines an emphasis on learning evidence and theory-based approaches to teaching and learning.

University of Montpellier

Montpellier, France

The 8th Surgical Robotics Summer School

September 2017

This summer school was a two-week intensive course in different areas of surgical robotics including the fundamental aspects, different applications and future trends.

VOLUNTEERING AND OUTREACH

- Mentor at the Fatima Fellowship program, which is an international mentorship program for aspiring researchers in computer science. It provides participants with research experience and guidance in preparing strong applications for PhD programs in computer science and machine learning. September 2024 - Present
- Mentor at Africa STEM Network (Afrisnet), a nonprofit organization that seeks to increase the overall participation of Africans in global and modern research in science, technology, engineering, and mathematics. August 2023 - Present
- Counselor at the counselling team at Egypt Scholars Inc. where we provide mentorship to current and prospective students regarding studying abroad and seeking financial support through scholarships. June 2016 - Present
- Judge at the Three Minute Thesis (3MT) Competition, Electrical and Computer Engineering Department, University of British Columbia February 2021
- Coordinator of the Pathways to Professorship program at the Faculty of Applied Science at UBC which aims to prepare PhD students for the rigors of applying to, and holding, an academic position. August 2019 - April 2020
- Cultural ambassador for the Vancouver Summer Program at the Faculty of Education at UBC. Summer 2018